The Super Spanning Connectivity of the Augmented Cubes

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Abstract

A *k-container* $C(u,v)$ of G between u and v is a set of k internally disjoint paths between u and v. A k-container $C(u,v)$ of G is a k^**-container* if it contains all nodes of G. A graph G is k^**-connected* if there exists a k^*-container between any two distinct nodes. The *spanning connectivity* of G, $\kappa^*(G)$, is defined to be the largest integer k such that G is w^*-connected for all $1 \leq w \leq k$ if G is a 1^*-connected graph and undefined if otherwise. A graph G is super spanning connected if $\kappa^*(G) = \kappa(G)$. In this paper, we prove that the n-dimensional augmented cube AQ_n is super spanning connected if and only if $n \neq 3$.

Keywords: hamiltonian, k-container, k-connected, hamiltonian connected.

1 Introduction

For the graph definition and notation we follow [2]. $G = (V,E)$ is a *graph* if V is a finite set and E is a subset of $\{(u,v) \mid (u,v)$ is an unordered pair of $V\}$. We say that V is the *node set* and E is the *edge set*. A graph H is a *subgraph* of graph G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. Two nodes u and v are *adjacent* if (u,v) is an edge of G. The set of *neighbors* of u, denote by $N_G(u)$, is $\{v \mid (u,v) \in E\}$. The *degree* $d_G(u)$ of a node u of G is the number of edges incident with u. A *path* is a sequence of nodes represented by $\langle v_0, v_1, \ldots, v_k \rangle$ with no repeated node and (v_i, v_{i+1}) is an edge of G for all $0 \leq i \leq k-1$. We also write the path $P = \langle v_0, v_1, \ldots, v_k \rangle$ as $\langle v_0, \ldots, v_i, Q, v_j, \ldots, v_k \rangle$, where Q is a path form v_i to v_j. And we use P^{-1} to denote the path $\langle v_k, v_{k-1}, \ldots, v_0 \rangle$. The *length* of a path Q, $l(Q)$, is the number of edges in Q. The *distance* $dist_G(u,v)$ of nodes u and v of G is the length of a shortest path joining u and v. A path is a *hamiltonian path* if it contains all nodes of G. A graph G is *hamiltonian connected* if there exists a hamiltonian path joining any two distinct nodes of G. A *cycle* is a path with at least three nodes such that the first node is the same as the last one. A *hamiltonian cycle* of G is a cycle that traverses every node of G. A graph is *hamiltonian* if it has a hamiltonian cycle.

A *k-container* $C(u,v)$ of G between u and v is a set of k internally disjoint paths between u and v. The concept of container is proposed by Hsu [8] to evaluate the performance of communication of an interconnection networks. The *connectivity* of G, $\kappa(G)$, is the minimum number of nodes whose removal leaves the remaining graph disconnected or trivial. It follows from Menger's Theorem [10] that there is a k-container between any two dis-

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tinct nodes of G if G is k-connected.

In this paper, we are interested in a specifies type of containers. A k-container $C(u,v)$ of G is a k^*-*container* if it contains all nodes of G. A graph G is k^*-*connected* if there exists a k^*-container between any two distinct nodes. A 1^*-connected graph is actually a hamiltonian connected graph. Moreover, a 2^*-connected graph is hamiltonian graph. Thus, the concept of k^*-connected graph is a hybrid concept of connectivity and hamiltonicity. The study of k^*-connected graph is motivated by the globally 3^*-connected graphs proposed by Albert, Aldred, and Holton [1]. A globally 3^*-connected graph is a cubic graph that is w^*-connected for all $1 \le w \le 3$. Recently, Lin et al [9] proved that the pancake graph P_n is w^*-connected for any w with $1 \le w \le n-1$ if and only if $n \ne 3$. Thus, we defined the *spanning connectivity* of a graph G, $\kappa^*(G)$, to be the largest integer k such that G is w^*-connected for all $1 \le w \le k$ if G is a 1^*-connected graph. A graph G is *super spanning connected* if $\kappa^*(G) = \kappa(G)$. Obviously, the complete graph K_n is super spanning connected. Moreover, P_n is super spanning connected if and only if $n \ne 3$.

In this paper, we prove that the n-dimensional augmented cube AQ_n is super spanning connected if and only if $n \ne 3$.

2 The augmented cubes

Let n be any positive integer. An n-bit binary string is a sequence $u_n u_{n-1} \dots u_1$ with $u_i \in \{0,1\}$ for each $1 \le i \le n$. Let $u = u_n u_{n-1} \dots u_1$ be an n-bit binary string. For $1 \le i \le n$, we set $(u)^i$ to denote the sequence $u_n u_{n-1} \dots u_{i+1} \overline{u}_i u_{i-1} \dots u_1$ and we set $(u)^{i-}$ to denote the sequence $u_n u_{n-1} \dots u_{i+1} \overline{u}_i \overline{u}_{i-1} \dots \overline{u}_1$. Obviously, $(u)^1 = (u)^{1-}$, $((u)^i)^i = u$, and $((u)^{i-})^{i-} = u$. The graph of the *n-dimensional augmented cube* [5], denoted by AQ_n, consists of all n-bit binary strings as its nodes. Two nodes $u = u_n u_{n-1} \dots u_1$ and $v = v_n v_{n-1} \dots v_1$ in AQ_n are adjacent if and only if there is $i \in \{1, 2, \dots, n\}$ such that either $(u)^i = v$ or $(u)^{i-} = v$.

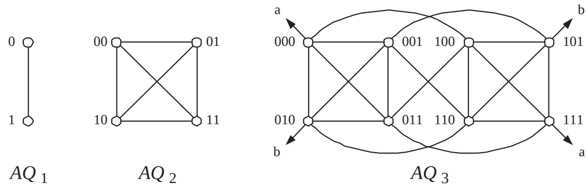

Figure 1: The augmented cubes AQ_1, AQ_1, and AQ_3.

The augmented cubes AQ_1, AQ_2, and AQ_3 are illustrated in Figure 1. It is proved in [5] that AQ_n is a node transitive, $(2n-1)$-regular, and $(2n-1)$-connected. We say that u_i is the i-th coordinate of u, denoted by $(u)_i$, for $1 \le i \le n$. For $i \in \{0,1\}$, let AQ_n^i denote the subgraph of AQ_n induced by those nodes u with $(u)_n = i$. For $n \ge 2$, AQ_n can be decomposed into 2 subgraph AQ_n^i, $0 \le i \le 1$, and each AQ_n^i is isomorphic to AQ_{n-1}. Thus, the augmented cube can be constructed recursively. For $0 \le i \le 1$ and $0 \le j \le 1$, let $AQ_n^{i,j}$ denote the subgraph of AQ_n induced by those nodes u with $(u)_n = i$ and $(u)_{n-1} = j$. For $n \ge 3$, AQ_n can be decomposed into 4 subgraph $AQ_n^{i,j}$, $0 \le i \le 1$ and $0 \le j \le 1$, and each $AQ_n^{i,j}$ is isomorphic to AQ_{n-2}.

Theorem 1 *[6] Let F be any subset of $V(AQ_n) \cup E(AQ_n)$. $AQ_n - F$ is hamiltonian if $|F| \le 2n-3$ and that $AQ_n - F$ is hamiltonian connected if $|F| \le 2n-4$ when $n = 2$ or $n \ge 4$. $AQ_3 - F$ is hamiltonian if $|F| \le 2$ and $AQ_3 - F$ is hamiltonian connected if $|F| \le 1$.*

Theorem 2 *[6] Let $n \ge 4$. Assume that $\{a,b\}$ and $\{c,d\}$ be two pairs of four distinct nodes of AQ_n. There exist two node-disjoint paths P_1 and P_2 such that (1) P_1 joins a and b, (2) P_2 joins c and d, and (3) $P_1 \cup P_2$ spans AQ_n.*

In Theorem 2, the nodes a, b, c, and d are distinct. Now, we let the nodes c and d are not distinct. By Theorem 1, we can find a hamiltonian path P between a and b in $AQ_n - \{c = d\}$. We state the same description in Theorem 2; let

$n \geq 4$, assume that nodes a, b, and c be three distinct nodes of AQ_n, node $d \neq a, b$, then there exist two node-disjoint paths P_1 and P_2 such that (1) P_1 joins a and b, (2) P_2 joins c and d, and (3) $P_1 \cup P_2$ spans AQ_n. We called the general form of Theorem 2 as property G2H.

Let f be a function on $V(AQ_n)$ and be defined by $f(u) = u$ if $(u)_n$=0 and $f(u) = (u)^{(n-1)-}$ if otherwise. The following Theorem can be proved easily.

Theorem 3 *The function f is an isomorphism of AQ_n into itself.*

3 The k^*-container of AQ_n

Lemma 1 *Let $n \geq 4$. Suppose that AQ_{n-1} is k^*-connected and u and v are two distinct nodes of AQ_n with $(u)_n = (v)_n$. Then there is a $(k+2)^*$-container of AQ_n between u and v.*

Proof. Without loss of generality, we assume that $(u)_n = 0$. Thus, $u \in AQ_n^0$ and $v \in AQ_n^0$. Let $\{P_1, P_2, \ldots, P_k\}$ be a k^*-container of AQ_n^0 between u and v with $l(P_k) \geq 3$.

Case 1: $v \neq (u)^{(n-1)-}$. Since AQ_n^1 is isomorphic to AQ_{n-1}. Let $\{(u)^n, (v)^{n-}\}$ and $\{(u)^{n-}, (v)^n\}$ be two pairs of four distinct nodes of AQ_n^1. By Theorem 2, there exist two node-disjoint paths Q_1 and Q_2 such that (1) Q_1 joins $(u)^n$ and $(v)^{n-}$, (2) Q_2 joins $(u)^{n-}$ and $(v)^n$, and (3) $Q_1 \cup Q_2$ spans AQ_n^1. We set $P'_{k+1} = \langle u, (u)^n, Q_1, (v)^{n-}, v\rangle$ and $P'_{k+2} = \langle u, (u)^{n-}, Q_2, (v)^n, v\rangle$. Then $\{P_1, P_2, \ldots, P_k, P'_{k+1}, P'_{k+2}\}$ forms a $(k+2)^*$-container of AQ_n between u and v.

Case 2: $v = (u)^{(n-1)-}$. Since $l(P_k) \geq 3$, we write $P_k = \langle u, i, j, P, v\rangle$ where i and j be the internal nodes of P_k. By Theorem 1, there is a hamiltonian path Q of $AQ_n^1 - \{(u)^n, (v)^n\}$ joining the nodes $(i)^n$ and $(j)^n$. We set $P'_k = \langle u, i, (i)^n, Q, (j)^n, j, P, v\rangle$, $P'_{k+1} = \langle u, (u)^n, v\rangle$, and $P'_{k+2} = \langle u, (u)^{n-}, v\rangle$. Then $\{P_1, P_2, \ldots, P_{k-1}, P'_k, P'_{k+1}, P'_{k+2}\}$ forms a $(k+2)^*$-container of AQ_n between u and v. □

Lemma 2 *Let $n \geq 2$. Suppose that AQ_{n-1} is k^*-connected and u and v are two adjacent nodes of AQ_n with $(u)_n \neq (v)_n$. Then there is a $(k+2)^*$-container of AQ_n between u and v.*

Proof. Without loss of generality, we assume that $(u)_n = 0$. By Theorem 3, we assume that $v = (u)^n$. Let $\{P_1, P_2, \ldots, P_k\}$ be a k^*-container of AQ_n^0 between u and $(v)^{n-}$ where $l(P_1) \geq l(P_2) \geq \cdots \geq l(P_k)$. Now, we write $P_i = \langle u = q_{t_i(1)}, q_{t_i(2)}, \ldots, q_{t_i(l(P_i))}, (v)^{n-}\rangle$ for $1 \leq i \leq k$. We have the following cases:

Case 1: $l(P_k) \geq 2$. We set $P'_i = \langle u = q_{t_i(1)}, q_{t_i(2)}, \ldots, q_{t_i(l(P_i))}, (q_{t_i(l(P_i))})^n, \ldots, (q_{t_i(2)})^n, (u)^n = v\rangle$ for $1 \leq i \leq k$, $P'_{k+1} = \langle u, (u)^{n-}, (v)^{n-}, v\rangle$, and $P'_{k+2} = \langle u, v\rangle$. Then $\{P'_1, P'_2, \ldots, P'_k, P'_{k+1}, P'_{k+2}\}$ forms a $(k+2)^*$-container of AQ_n between u and v.

Case 2: $l(P_k) = 1$. We set $P'_i = \langle u = q_{t_i(1)}, q_{t_i(2)}, \ldots, q_{t_i(l(P_i))}, (q_{t_i(l(P_i))})^n, \ldots, (q_{t_i(2)})^n, (u)^n = v\rangle$ for $1 \leq i \leq k-1$, $P'_k = \langle u, (v)^{n-}, v\rangle$, $P'_{k+1} = \langle u, (u)^{n-}, v\rangle$, and $P'_{k+2} = \langle u, v\rangle$. Then $\{P'_1, P'_2, \ldots, P'_k, P'_{k+1}, P'_{k+2}\}$ forms a $(k+2)^*$-container of AQ_n between u and v. □

Lemma 3 *Let k and n be any two positive integer with $3 \leq k \leq 2n-4$. Suppose that u and v are two not adjacent nodes of AQ_n with $(u)_n \neq (v)_n$. Then there is a $(k+1)^*$-container of AQ_n between u and v if there is a k^*-container between any two nodes of AQ_n^i for $0 \leq i \leq 1$.*

Proof. Obviously, $n \geq 4$. Without loss of generality, we assume that $(u)_n = 0$, $(v)_n = 1$, and $v \neq (u)^n, (u)^{n-}$. Let $\{Q_1, Q_2, \ldots, Q_k\}$ be a k^*-container of AQ_n^0 between u and $(v)^n$. We write $Q_i = \langle u, H_i, x_i, (v)^n\rangle$ for every $1 \leq i \leq k$. Since $k \geq 3$, we assume that $l(H_i) \geq 1$ for every $2 \leq i \leq k$. Let $F = \{(x_3)^n, (x_4)^n, \ldots, (x_k)^n\}$. Since $k-2 \leq 2n-6$, by Theorem 1, there is a hamiltonian path R of $AQ_n^1 - F$ joining $(u)^n$ to $(x_2)^n$. We write R as $\langle (u)^n, R_1, v, R_2, (x_2)^n\rangle$. We set

$$P_1 = \langle u, H_1, x_1, (v)^n, v\rangle,$$
$$P_2 = \langle u, H_2, x_2, (x_2)^n, R_2^{-1}, v\rangle,$$

$$P_i = \langle u, H_i, x_i, (x_i)^n, v\rangle \text{ for every } 3 \le i \le k,$$
$$P_{k+1} = \langle u, (u)^n, R_1, v\rangle.$$

Then $\{P_1, P_2, \ldots, P_{k+1}\}$ forms a $(k+1)^*$-container of AQ_n between u and v. $\square$

Lemma 4 *Let $n \ge 4$. Suppose that x and y are two nodes of AQ_n with $dist_{AQ_n}(x, y) = 2$. Then $\{(x)^{n-1}, (x)^{(n-1)-}\} \cap \{(y)^{n-1}, (y)^{(n-1)-}\} = \emptyset$.*

Proof. We prove this statement by contradiction. Let $x = x_n x_{n-1} \ldots x_1$, $y = y_n y_{n-1} \ldots y_1$. If $(x)^{n-1} = (y)^{n-1}$, then $x = y$. If $(x)^{n-1} = (y)^{(n-1)-}$, then $x_n \overline{x}_{n-1}\ x_{n-2} \ldots x_1 = y_n \overline{y}_{n-1}\ \overline{y}_{n-2} \cdots \overline{y}_1$. We implies that $dist_{AQ_n}(x, y) = 1$. It is a contradiction. Then node $(x)^{n-1}$ is not the same as the nodes $(y)^{n-1}$ or $(y)^{(n-1)-}$. By the same reason, node $(x)^{(n-1)-}$ is not the same as the nodes $(y)^{n-1}$ or $(y)^{(n-1)-}$. $\square$

Lemma 5 *AQ_n is 3^*-connected if $n \ge 2$.*

Proof. Since AQ_2 is isomorphic to complete graph K_4 with 4 nodes, AQ_2 is 3^*-connected. Thus, we assume that $n \ge 3$. Let u and v be any two distinct nodes of AQ_n. We need to find a 3^*-container of AQ_n between u and v. Without loss of generality, we assume that $u \in AQ_n^0$.

Suppose that $v \in AQ_n^0$. By Theorem 1, there is a 2^*-container $\{P_1, P_2\}$ of AQ_n^0 between u and v. Again, there is a hamiltonian path H of AQ_n^1 joining $(u)^n$ to $(v)^n$. We set $P_3 = \langle u, (u)^n, H, (v)^n, v\rangle$. Then $\{P_1, P_2, P_3\}$ forms a 3^*-container of AQ_n between u and v.

Suppose that $v \in AQ_n^1$. Obviously, either $(u)^n \ne v$ or $(u)^{n-} \ne v$. Without loss of generality, we assume that $(u)^n \ne v$. Note that $(v)^n \ne u$. Since $n \ge 3$, $|V(AQ_n^0)| \ge 4$. We can choose a node x in $AQ_n^0 - \{u, (v)^n\}$. By Theorem 1, there is a hamiltonian path H of AQ_n^0 joining x to $(v)^n$. Again, there is a hamiltonian path R of AQ_n^1 joining $(u)^n$ to $(x)^n$. Without loss of generality, we write $H = \langle x, H_1, u, H_2, (v)^n\rangle$ and $R = \langle (u)^n, R_1, v, R_2, (x)^n\rangle$. We set

$$P_1 = \langle u, (u)^n, R_1, v\rangle,$$
$$P_2 = \langle u, H_1^{-1}, x, (x)^n, R_2^{-1}, v\rangle, \text{ and}$$
$$P_3 = \langle u, H_2, (v)^n, v\rangle.$$

Then $\{P_1, P_2, P_3\}$ forms a 3^*-container of AQ_n between u and v.

Thus, this statement is proved. $\square$

Lemma 6 *AQ_3 is k^*-connected for every $1 \le k \le 4$ and AQ_3 is not 5^*-connected. Hence, AQ_3 is not super spanning connected.*

Proof. By Theorem 1, AQ_3 is 1^*-connected and 2^*-connected. By Lemma 5, AQ_3 is 3^*-connected. Let u and v be any two distinct nodes of AQ_n. Since AQ_3 is node transitive, we set that $u = 000$. By Theorem 3, we assume that $v \in \{001, 011, 100, 101\}$. With the following table, we can prove that AQ_3 is 4^*-connected.

$000 \to 001$	(000, 011, 001)(000, 010, 001)(000, 100, 101, 001)(000, 111, 110, 001)
$000 \to 011$	(000, 011)(000, 001, 011)(000, 010, 011)(000, 100, 101, 110, 111, 011)
$000 \to 100$	(000, 100)(000, 001, 101, 100)(000, 011, 111, 100)(000, 010, 110, 100)
$000 \to 101$	(000, 001, 101)(000, 100, 101)(000, 010, 101)(000, 011, 111, 110, 101)

However, let $u = 000$ and $v = 011$. Then, $N_{AQ_3}(u) \cap N_{AQ_3}(v) = \{001, 010, 100, 111\}$. Obviously, $\{P_1 = \langle 000, 001, 011\rangle$, $P_2 = \langle 000, 010, 011\rangle$, $P_3 = \langle 000, 100, 011\rangle$, $P_4 = \langle 000, 111, 011\rangle$, $P_5 = \langle 000, 011\rangle\}$ is the only possible 5-container between u and v. However, $\{P_1, P_2, \ldots, P_5\}$ is not a 5^*-container of AQ_3 between u and v. Hence, AQ_3 is not 5^*-connected. $\square$

Lemma 7 *AQ_4 is super spanning connected.*

Proof. By Theorem 1 and Lemma 5, AQ_4 is 1^*-connected, 2^*-connected, and 3^*-connected. Let u and v be any two distinct nodes in AQ_4. Since AQ_4 is node transitive, we assume that $u = 0000$. Now, we need to show there is a k^*-container of AQ_4 between u and v for every $4 \le k \le 7$. We have the following cases.

Case 1. $v \in AQ_4^0$. Since AQ_4^0 is isomorphic to AQ_3, by Lemmas 1 and 6, there is a k^*-container of AQ_4 between u and v for $4 \le k \le 6$. Thus, we consider that $k = 7$. The 7^*-container of AQ_4 between $u = 0000$ and $v \in \{0001, 0010, 0011, 0100, 0101, 0110, 0111\}$ are list below.

v	$7^* - container$
0001	$\langle 0000, 0001\rangle$ $\langle 0000, 0010, 0001\rangle$ $\langle 0000, 0011, 0001\rangle$ $\langle 0000, 0100, 0101, 0001\rangle$ $\langle 0000, 0111, 0110, 0001\rangle$ $\langle 0000, 1000, 1010, 1011, 1001, 0001\rangle$ $\langle 0000, 1111, 1101, 1100, 1110, 0001\rangle$
0010	$\langle 0000, 0010\rangle$ $\langle 0000, 0001, 0010\rangle$ $\langle 0000, 0011, 0010\rangle$ $\langle 0000, 0100, 0110, 0010\rangle$ $\langle 0000, 0101, 0111, 0010\rangle$ $\langle 0000, 1000, 1001, 1011, 1010, 0010\rangle$ $\langle 0000, 1111, 1110, 1100, 1101, 0010\rangle$
0011	$\langle 0000, 0011\rangle$ $\langle 0000, 0001, 0011\rangle$ $\langle 0000, 0010, 0011\rangle$ $\langle 0000, 0100, 0011\rangle$ $\langle 0000, 0111, 0011\rangle$ $\langle 0000, 1000, 1001, 1010, 1011, 0011\rangle$ $\langle 0000, 1111, 1110, 0110, 0101, 1101, 1100, 0011\rangle$
0100	$\langle 0000, 0100\rangle$ $\langle 0000, 0111, 0100\rangle$ $\langle 0000, 0011, 0100\rangle$ $\langle 0000, 0010, 0110, 0100\rangle$ $\langle 0000, 0001, 0101, 0100\rangle$ $\langle 0000, 1000, 1001, 1010, 1011, 0100\rangle$ $\langle 0000, 1111, 1110, 1101, 1100, 0100\rangle$
0101	$\langle 0000, 0001, 0101\rangle$ $\langle 0000, 0010, 0101\rangle$ $\langle 0000, 0100, 0101\rangle$ $\langle 0000, 0111, 0101\rangle$ $\langle 0000, 1111, 1101, 0101\rangle$ $\langle 0000, 0011, 1100, 1110, 0110, 0101\rangle$ $\langle 0000, 1000, 1001, 1011, 1010, 0101\rangle$
0110	$\langle 0000, 0001, 0110\rangle$ $\langle 0000, 0010, 0110\rangle$ $\langle 0000, 0100, 0110\rangle$ $\langle 0000, 0111, 0110\rangle$ $\langle 0000, 1000, 1001, 0110\rangle$ $\langle 0000, 0011, 1011, 1010, 0101, 0110\rangle$ $\langle 0000, 1111, 1101, 1100, 1110, 0110\rangle$
0111	$\langle 0000, 0111\rangle$ $\langle 0000, 0011, 0111\rangle$ $\langle 0000, 0100, 0111\rangle$ $\langle 0000, 1000, 0111\rangle$ $\langle 0000, 1111, 0111\rangle$ $\langle 0000, 0011, 1010, 1101, 0101, 0111\rangle$ $\langle 0000, 0001, 1001, 1011, 1100, 1110, 0110, 0111\rangle$

Case 2. $v \in AQ_4^1$ and $v \in \{1000, 1111\}$. By Theorem 3, we assume that $v = 1000$. By Lemmas 2 and 6, there is a k^*-container of AQ_4 between u and v for $4 \leq k \leq 6$. Thus, we consider that $k = 7$. The 7^*-container of AQ_4 between $u = 0000$ and $v = 1000$ is list below.

v	$7^* - container$
1000	$\langle 0000, 1000\rangle$ $\langle 0000, 0111, 1000\rangle$ $\langle 0000, 1111, 1000\rangle$ $\langle 0000, 0100, 1100, 1000\rangle$ $\langle 0000, 0011, 1011, 1000\rangle$ $\langle 0000, 0010, 0110, 1110, 1010, 1000\rangle$ $\langle 0000, 0001, 0101, 1101, 1001, 1000\rangle$

Case 3. $v \in AQ_4^1$ and $v \notin \{1000, 1111\}$. By Theorem 3, we assume that $v \in \{1001, 1010, 1011\}$. By Lemmas 3 and 6, there is a k^*-container of AQ_4 for $4 \leq k \leq 5$ between u and v. Thus, we consider that $k = 6, 7$. The 6^*-container and the 7^*-container of AQ_4 between $u = 0000$ and $v \in \{1001, 1010, 1011\}$ are list below.

v	$6^* - container$
1001	$\langle 0000, 0010, 1101, 1001\rangle$ $\langle 0000, 0100, 1011, 1001\rangle$ $\langle 0000, 0111, 0110, 1001\rangle$ $\langle 0000, 1111, 1110, 1001\rangle$ $\langle 0000, 0001, 0101, 1010, 1001\rangle$ $\langle 0000, 0011, 1100, 1000, 1001\rangle$
1010	$\langle 0000, 0001, 1001, 1010\rangle$ $\langle 0000, 0011, 1011, 1010\rangle$ $\langle 0000, 0111, 0101, 1010\rangle$ $\langle 0000, 1111, 1101, 1010\rangle$ $\langle 0000, 0010, 0110, 1110, 1010\rangle$ $\langle 0000, 0100, 1100, 1000, 1010\rangle$
1011	$\langle 0000, 0011, 1011\rangle$ $\langle 0000, 1000, 1011\rangle$ $\langle 0000, 0001, 1110, 1111, 1011\rangle$ $\langle 0000, 0010, 1101, 1100, 1011\rangle$ $\langle 0000, 0100, 0101, 1010, 1011\rangle$ $\langle 0000, 0111, 0110, 1001, 1011\rangle$

v	$7^* - container$
1001	$\langle 0000, 0001, 1001\rangle$ $\langle 0000, 1000, 1001\rangle$ $\langle 0000, 0010, 1010, 1001\rangle$ $\langle 0000, 1111, 1011, 1001\rangle$ $\langle 0000, 0111, 0110, 1001\rangle$ $\langle 0000, 0011, 1100, 1110, 1001\rangle$ $\langle 0000, 0100, 0101, 1101, 1001\rangle$
1010	$\langle 0000, 0010, 1010\rangle$ $\langle 0000, 1000, 1010\rangle$ $\langle 0000, 0001, 1001, 1010\rangle$ $\langle 0000, 0011, 1011, 1010\rangle$ $\langle 0000, 1111, 1110, 1010\rangle$ $\langle 0000, 0100, 1100, 1101, 1010\rangle$ $\langle 0000, 0111, 0110, 0101, 1010\rangle$
1011	$\langle 0000, 0011, 1011\rangle$ $\langle 0000, 0100, 1011\rangle$ $\langle 0000, 1000, 1011\rangle$ $\langle 0000, 1111, 1011\rangle$ $\langle 0000, 0001, 0101, 1010, 1011\rangle$ $\langle 0000, 0010, 1101, 1001, 1011\rangle$ $\langle 0000, 0111, 0110, 1110, 1100, 1011\rangle$

□

Lemma 8 *AQ_5 is super spanning connected.*

Proof. By Theorem 1 and Lemma 5, AQ_5 is 1^*-connected, 2^*-connected, and 3^*-connected. Let u and v be any two distinct nodes in AQ_5. Since AQ_5 is node transitive, we assume that $u = 00000$. Obviously, $u \in AQ_5^0$. We need to show there is a k^*-container of AQ_5 between u and v for every $4 \leq k \leq 9$. We have the following cases.

Case 1. $v \in AQ_5^0$. Since AQ_5^0 is isomorphic to AQ_4, by Lemmas 1 and 7, there is a k^*-container of AQ_5 between u and v for every $4 \leq k \leq 9$.

Case 2. $v \in AQ_5^1$ and $v \in \{10000, 11111\}$. By Lemmas 2 and 7, there is a k^*-container of AQ_5 between u and v for every $4 \leq k \leq 9$.

Case 3. $v \in AQ_5^1$ and $v \notin \{10000, 11111\}$. By Theorem 3, we assume that $v \in \{10001, 10010, 10011, 10100, 10101, 10110, 10111\}$. By Lemmas 3 and 7, there is a k^*-container of AQ_5 for $4 \leq k \leq 7$ between u and v. The 8^*-container and the 9^*-container of AQ_5 between $u = 00000$ and $v \in \{10001, 10010, 10011, 10100, 10101, 10110, 10111\}$ are list below.

v	$8^* - container$
10001	$\langle 00000, 01000, 01001, 01011, 01010, 11010, 11000, 11011, 11001, 10001\rangle$ $\langle 00000, 01111, 01101, 01100, 01110, 10001\rangle$ $\langle 00000, 11111, 11101, 11100, 11110, 10001\rangle$ $\langle 00000, 00001, 10001\rangle$ $\langle 00000, 10000, 10001\rangle$ $\langle 00000, 00011, 10011, 10001\rangle$ $\langle 00000, 00010, 10010, 10001\rangle$ $\langle 00000, 00100, 00110, 00101, 00111, 10111, 10101, 10100, 10110, 10001\rangle$
10010	$\langle 00000, 01000, 01010, 01011, 01001, 11001, 11011, 11000, 11010, 10010\rangle$ $\langle 00000, 01111, 01110, 01100, 01101, 10010\rangle$ $\langle 00000, 11111, 11110, 11100, 11101, 10010\rangle$ $\langle 00000, 00010, 10010\rangle$ $\langle 00000, 10000, 10010\rangle$ $\langle 00000, 00001, 10001, 10010\rangle$ $\langle 00000, 00011, 10011, 10010\rangle$ $\langle 00000, 00100, 00110, 00111, 00101, 10101, 10111, 10100, 10110, 10010\rangle$
10011	$\langle 00000, 01000, 01001, 01011, 01010, 11010, 11000, 11001,$

	11011, 10011⟩ ⟨00000, 01111, 01101, 01110, 01100, 10011⟩ ⟨00000, 11111, 11101, 11110, 11100, 10011⟩ ⟨00000, 10000, 10011⟩ ⟨00000, 00001, 10001, 10011⟩ ⟨00000, 00010, 10010, 10011⟩ ⟨00000, 00011, 10011⟩ ⟨00000, 00100, 00110, 00111, 00101, 10101, 10111, 10110, 10100, 10011⟩
10101	⟨00000, 01000, 11000, 11001, 11011, 11010, 10101⟩ ⟨00000, 01111, 01101, 01100, 01110, 01001, 01011, 01010, 10101⟩ ⟨00000, 11111, 11110, 11100, 11101, 10101⟩ ⟨00000, 00001, 10001, 10101⟩ ⟨00000, 00010, 10010, 10010, 10101⟩ ⟨00000, 10000, 10100, 10101⟩ ⟨00000, 00011, 10011, 10111, 10101⟩ ⟨00000, 00100, 00101, 00111, 00110, 10110, 10101⟩
10100	⟨00000, 01000, 01010, 01001, 01011, 10100⟩ ⟨00000, 01111, 01101, 01110, 01100, 11100, 10100⟩ ⟨00000, 11111, 11101, 11110, 11001, 11000, 11010, 11011, 10100⟩ ⟨00000, 10000, 10100⟩ ⟨00000, 00011, 10011, 10100⟩ ⟨00000, 00001, 10001, 10101, 10100⟩ ⟨00000, 00010, 10010, 10110, 10100⟩ ⟨00000, 00100, 00101, 00110, 00111, 10111, 10100⟩
10110	⟨00000, 01000, 01011, 11011, 11010, 11000, 11001, 10110⟩ ⟨00000, 01111, 01101, 01100, 01110, 01010, 01001, 10110⟩ ⟨00000, 11111, 11101, 11100, 11110, 10110⟩ ⟨00000, 00001, 10001, 10110⟩ ⟨00000, 00010, 10010, 10110⟩ ⟨00000, 00011, 10011, 10110⟩ ⟨00000, 10000, 10100, 10110⟩ ⟨00000, 00100, 00110, 00111, 00101, 10101, 10110⟩
10111	⟨00000, 11111, 10111⟩ ⟨00000, 01000, 10111⟩ ⟨00000, 10000, 10111⟩ ⟨00000, 00111, 10111⟩ ⟨00000, 00011, 10011, 10111⟩ ⟨00000, 00010, 01010, 01011, 01001, 11001, 11011, 11010, 11000, 10111⟩ ⟨00000, 01111, 01101, 01100, 01110, 11110, 11100, 11101, 10010, 10001, 10101, 10111⟩ ⟨00000, 00001, 00101, 00100, 00110, 10110, 10100, 10111⟩
v	$9^* - container$
10001	⟨00000, 01000, 01001, 01011, 01010, 11010, 11000, 11011, 11001, 10001⟩ ⟨00000, 01111, 01101, 01100, 01110, 10001⟩ ⟨00000, 11111, 11101, 11100, 11110, 10001⟩ ⟨00000, 00001, 10001⟩ ⟨00000, 10000, 10001⟩ ⟨00000, 00011, 10011, 10001⟩ ⟨00000, 00010, 10010, 10001⟩ ⟨00000, 00100, 00110, 00101, 10101, 10001⟩ ⟨00000, 00111, 10111, 10100, 10110, 10001⟩
10010	⟨00000, 01000, 01010, 01011, 01001, 11001, 11011, 11000, 11010, 10010⟩ ⟨00000, 01111, 01110, 01100, 01101, 10010⟩ ⟨00000, 11111, 11110, 11100, 11101, 10010⟩ ⟨00000, 00010, 10010⟩ ⟨00000, 10000, 10010⟩ ⟨00000, 00001, 10001, 10010⟩ ⟨00000, 00011, 10011, 10010⟩ ⟨00000, 00100, 00110, 00101, 10101, 10010⟩ ⟨00000, 00111, 10111, 10100, 10110, 10010⟩
10011	⟨00000, 01000, 01001, 01011, 01010, 11010, 11000, 11001, 11011, 10011⟩ ⟨00000, 01111, 01101, 01110, 01100, 10011⟩ ⟨00000, 11111, 11101, 11110, 11100, 10011⟩ ⟨00000, 10000, 10011⟩ ⟨00000, 00001, 10001, 10011⟩ ⟨00000, 00010, 10010, 10011⟩ ⟨00000, 00011, 10011⟩ ⟨00000, 00100, 00110, 10110, 10100, 10011⟩ ⟨00000, 00111, 00101, 10101, 10111, 10011⟩
10101	⟨00000, 01000, 11000, 11001, 11011, 11010, 10101⟩ ⟨00000, 01111, 01101, 01100, 01110, 01001, 01011, 01010, 10101⟩ ⟨00000, 11111, 11110, 11100, 11101, 10101⟩ ⟨00000, 00001, 10001, 10101⟩ ⟨00000, 00010, 10010, 10010, 10101⟩ ⟨00000, 10000, 10100, 10101⟩ ⟨00000, 00011, 10011, 10111, 10101⟩ ⟨00000, 00100, 00101, 10101⟩ ⟨00000, 00111, 00110, 10101⟩
10100	⟨00000, 01000, 01010, 01001, 01011, 10100⟩ ⟨00000, 01111, 01101, 00101, 00110, 01110, 01100, 11100, 10100⟩ ⟨00000, 11111, 11101, 11110, 11001, 11000, 11010, 11011, 10100⟩ ⟨00000, 10000, 10100⟩ ⟨00000, 00011, 10011, 10100⟩ ⟨00000, 00001, 10001, 10101, 10100⟩ ⟨00000, 00010, 10010, 10110, 10100⟩ ⟨00000, 00111, 10111, 10100⟩ ⟨00000, 00100, 10100⟩
10110	⟨00000, 01000, 01011, 11011, 11010, 11000, 11001, 10110⟩ ⟨00000, 01111, 01101, 01100, 01110, 01010, 01001, 10110⟩ ⟨00000, 11111, 11101, 11100, 11110, 10110⟩ ⟨00000, 00001, 10001, 10110⟩ ⟨00000, 00010, 10010, 10110⟩ ⟨00000, 00011, 10011, 10110⟩ ⟨00000, 10000, 10100, 10110⟩ ⟨00000, 00111, 00101, 10101, 10110⟩ ⟨00000, 00100, 00110, 10110⟩
10111	⟨00000, 11111, 10111⟩ ⟨00000, 01000, 10111⟩ ⟨00000, 10000, 10111⟩ ⟨00000, 00111, 10111⟩ ⟨00000, 00011, 10011, 10111⟩ ⟨00000, 00010, 01010, 01011, 01001, 11001, 11011, 11010, 11000, 10111⟩ ⟨00000, 01111, 01101, 01100, 01110, 11110, 11100, 11101, 10010, 10001, 10101, 10111⟩ ⟨00000, 00001, 00101, 00110, 10110, 10111⟩ ⟨00000, 00100, 10100, 10111⟩

□

Lemma 9 *AQ_n is super spanning connected if $n > 3$.*

Proof. Let u and v be any two distinct nodes in AQ_n. By Theorem 1 and Lemma 5, there are 1^*-container, 2^*-container, and 3^*-container of AQ_n between u and v. We prove the remaining case, k^*-container for $4 \leq k \leq 2n-1$, of this statement by induction on n. By Lemmas 7 and 8, this statement holds on AQ_4 and AQ_5. Thus, we assume this statement holds on AQ_t for every $4 \leq t < n$.

By Theorem 1 and Lemma 1, there is a k^*-container of AQ_n between u and v when $(u)_n = (v)_n$ for every $1 \leq k \leq 2n-1$.

By Theorem 1 and Lemma 2, there is a k^*-container of AQ_n between any two adjacent nodes u and v when $(u)_n \neq (v)_n$ for every $1 \leq k \leq 2n-1$.

Now, we consider the remaining cases that nodes u and v are not adjacent and $(u)_n \neq (v)_n$. Since AQ_n is node transitive, we assume that $u \in AQ_n^{0,0}$. By Theorem 3, we assume that $v \in AQ_n^{1,0}$. Now, we consider the k^*-container of AQ_n for $4 \leq k \leq 2n-1$ in the following cases.

Case I: $4 \leq k \leq 2n-3$. By induction hypothesis, there is a k^*-container of AQ_n^0 and AQ_n^1 between any two nodes for $3 \leq k \leq 2n-4$. By Lemma 3, there is a k^*-container of AQ_n between u and v for $4 \leq k \leq 2n-3$.

Case II: $k = 2n-2$.

Case 1: nodes u and $(v)^n$ are not adjacent. By induction hypothesis, let $\{\langle u, U_1, w_1, (v)^n\rangle, \langle u, U_2, w_2, (v)^n\rangle, \ldots, \langle u, U_{2n-5}, w_{2n-5}, (v)^n\rangle\}$ be a $(2n-5)^*$-container of $AQ_n^{0,0}$ between u and $(v)^n$. Without lose of generality, we let $w_1 = ((v)^n)^2$, $w_2 = ((v)^n)^3$, and $w_3 = ((v)^n)^4$. Hence,

$dist_{AQ_n^{0,0}}(w_i, w_j) \geq 2$, $dist_{AQ_n^{1,0}}((w_i)^n, (w_j)^n) \geq 2$ for $1 \leq i, j \leq 3$ and $i \neq j$. We set

$$\begin{aligned} P_1 &= \langle u, U_1, w_1, (v)^n, v\rangle, \text{ and} \\ P_i &= \langle u, U_i, w_i, (w_i)^n, v\rangle \text{ for every } 2 \leq i \leq 2n-5. \end{aligned}$$

Now, we construct the remaining paths. By Theorem 1, there is a hamiltonian path H of $AQ_n^{1,0} - \{(w_j)^n | 4 \leq j \leq 2n-5\}$ between $(u)^n$ and $(w_3)^n$. We consider the forms of hamiltonian path H in the following cases:

Subcase 1.1: $H = \langle (u)^n, R, (w_1)^n, S, (w_2)^n, v, (w_3)^n\rangle$ where $S = \langle s_0 = (w_1)^n, s_1, \ldots, s_j = (w_2)^n\rangle$. We know that $j \geq 2$, because $dist_{AQ_n^{1,0}}((w_1)^n, (w_2)^n) \geq 2$. And the neighbors of v in $AQ_n^{1,0} - \{(w_j)^n |\ 4 \leq j \leq 2n-5\}$ is the nodes $(w_1)^n$, $(w_2)^n$, and $(w_3)^n$. Hence, $dist_{AQ_n^{1,0}}(s_{j-1}, v) = 2$. By Lemma 4, $\{(s_{j-1})^{n-1}, (s_{j-1})^{(n-1)-}\} \cap \{(v)^{n-1}, (v)^{(n-1)-}\} = \emptyset$. We can find at least one node q in $\{(s_{j-1})^{n-1}, (s_{j-1})^{(n-1)-}\}$ such that $q \notin \{(u)^{n-}, (v)^{n-1}, (v)^{(n-1)-}\}$. Without loss of generality, we let $q = (s_{j-1})^{n-1}$. By Theorem 1, there is a hamiltonian path T of $AQ_n^{1,1}$ between $(s_{j-1})^{n-1}$ and $(v)^{n-1}$. And now, we consider the paths in $AQ_n^{0,1}$. It is easy to know that $(s_1)^{n-} \neq (v)^{n-}$. We set $g_1 = (u)^{n-1}$ and $g_2 = (u)^{(n-1)-}$ if $(s_1)^{n-} = (u)^{(n-1)-}$, $g_1 = (u)^{(n-1)-}$ and $g_2 = (u)^{n-1}$ if otherwise. By property G2H, there exist two node-disjoint paths X_1 and X_2 such that (1) X_1 joins g_1 and $(v)^{n-}$, (2) X_2 joins g_2 and $(s_1)^{n-}$, and (3) $X_1 \cup X_2$ spans $AQ_n^{0,1}$. We set

$$\begin{aligned} P_{2n-4} &= \langle u, (u)^n, R, (w_1)^n, v\rangle, \\ P_{2n-3} &= \langle u, g_1, X_1, (v)^{n-}, v\rangle, \text{ and} \\ P_{2n-2} &= \langle u, g_2, X_2, (s_1)^{n-}, s_1, \ldots, s_{j-1}, \\ &\quad (s_{j-1})^{n-1}, T, (v)^{n-1}, v\rangle. \end{aligned}$$

See Figure 2 for an illustration.

Subcase 1.2: $H = \langle (u)^n, R, (w_1)^n, v, (w_2)^n, S, (w_3)^n\rangle$ where $S = \langle s_0 = (w_2)^n, s_1, \ldots, s_j = (w_3)^n\rangle$. The same idea considered in Subcase 1.1. We set

$$\begin{aligned} P_{2n-4} &= \langle u, (u)^n, R, (w_1)^n, v\rangle, \\ P_{2n-3} &= \langle u, g_1, X_1, (v)^{n-}, v\rangle, \text{ and} \end{aligned}$$

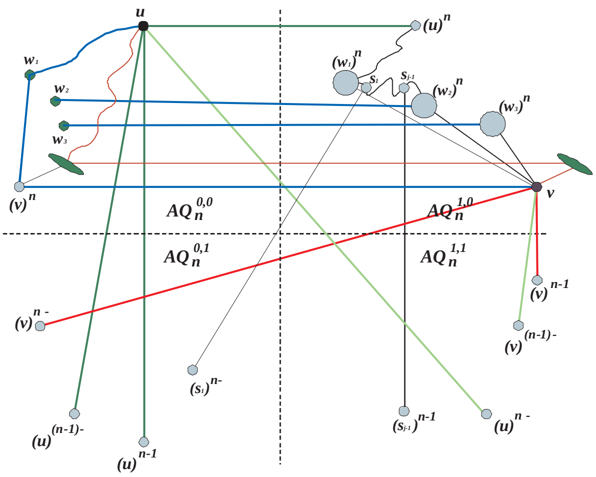

Figure 2: Illustration for Lemma 9, Subcase 1.1.

$$\begin{aligned} P_{2n-2} &= \langle u, g_2, X_2, (s_1)^{n-}, s_1, \ldots, s_{j-1}, \\ &\quad (s_{j-1})^{n-1}, T, (v)^{n-1}, v\rangle. \end{aligned}$$

Subcase 1.3: $H = \langle (u)^n, S, (w_2)^n, R, (w_1)^n, v, (w_3)^n\rangle$ where $S = \langle s_0 = (w_2)^n, s_1, \ldots, s_j = (w_1)^n\rangle$ and $R = \langle r_0 = (u)^n, r_1, \ldots, r_i = (w_1)^n\rangle$. The same idea considered in Subcase 1.1., we can find g_1 and g_2 in $AQ_n^{0,1}$. And there is a hamiltonian path T of $AQ_n^{1,1}$ between $(s_{j-1})^{n-1}$ and $(v)^{n-1}$. We set

$$\begin{aligned} P_{2n-4} &= \langle u, g_1, X_1, (v)^{n-}, v\rangle, \\ P_{2n-3} &= \langle u, g_2, X_2, (r_1)^{n-}, r_1, \ldots, r_{i-1}, (w_1)^n, v\rangle, \text{ and} \\ P_{2n-2} &= \langle u, (u)^n = s_0, \ldots, s_{j-1}, (s_{j-1})^{n-1}, T, \\ &\quad (v)^{n-1}, v\rangle. \end{aligned}$$

Subcase 1.4: $H = \langle (u)^n, S, (w_2)^n, v, (w_1)^n, R, (w_3)^n\rangle$ where $S = \langle s_0 = (w_1)^n, s_1, \ldots, s_j = (w_3)^n\rangle$ and $R = \langle r_0 = (u)^n, r_1, \ldots, r_i = (w_1)^n\rangle$. The same idea considered in Subcase 1.3. We set

$$\begin{aligned} P_{2n-4} &= \langle u, g_1, X_1, (v)^{n-}, v\rangle, \\ P_{2n-3} &= \langle u, g_2, X_2, (r_{i-1})^{n-}, r_{i-1}, \ldots, r_1, (w_1)^n, v \text{ and } \rangle, \\ P_{2n-2} &= \langle u, (u)^n = s_0, \ldots, s_{j-1}, (s_{j-1})^{n-1}, T, \\ &\quad (v)^{n-1}, v\rangle. \end{aligned}$$

Case 2: Nodes u and $(v)^n$ are adjacent. By induction hypothesis, let $\{\langle u, (v)^n\rangle, \langle u, U_2, w_2, (v)^n\rangle,$

$\ldots, \langle u, U_{2n-5}, w_{2n-5}, (v)^n \rangle\}$ be a $(2n-5)^*$-container of $AQ_n^{0,0}$ between u and $(v)^n$. We set

$$\begin{aligned} P_1 &= \langle u, (v)^n, v \rangle, \\ P_i &= \langle u, U_i, w_i, (w_i)^n, v \rangle \text{ for every } 2 \leq i \leq 2n-5, \text{ and} \\ P_{2n-4} &= \langle u, (u)^n, v \rangle. \end{aligned}$$

Now, we construct the remaining paths. By Theorem 1, there is a hamiltonian path H of $AQ_n^{1,0} - \{(w_j)^n | 4 \leq j \leq 2n-5\}$ between $(u)^n$ and $(w_3)^n$. We consider the forms of hamiltonian path H in the following cases:

Subcase 2.1: $H = \langle (u)^n, v, (w_2)^n, S, (w_3)^n \rangle$ where $S = \langle s_0 = (w_2)^n, s_1, \ldots, s_j = (w_3)^n \rangle$. Because $|AQ_n^{1,0}| - |\{(u)^n, (w_2)^n, \ldots, (w_{2n-5})^n\}| \geq 2$ when $n \geq 6$. Hence, $j \geq 3$. By Lemma 4, $\{(s_{j-1})^{n-1}, (s_{j-1})^{(n-1)-}\} \cap \{(v)^{n-1}, (v)^{(n-1)-}\} = \emptyset$. We can find at least one node q in $\{(s_{j-1})^{n-1}, (s_{j-1})^{(n-1)-}\}$ such that $q \notin \{(u)^{n-}, (v)^{n-1}, (v)^{(n-1)-}\}$. Without loss of generality, we let $q = (s_{j-1})^{n-1}$. By Theorem 1, there is a hamiltonian path T of $AQ_n^{1,1}$ between $(s_{j-1})^{n-1}$ and $(v)^{n-1}$. And now, we consider the paths in $AQ_n^{0,1}$. It is easy to know that $(s_1)^{n-} \neq (v)^{n-}$. We set $g_1 = (u)^{n-1}$ and $g_2 = (u)^{(n-1)-}$ if $(s_1)^{n-} = (u)^{(n-1)-}$, $g_1 = (u)^{(n-1)-}$ and $g_2 = (u)^{n-1}$ if otherwise. By property G2H, there exist two node-disjoint paths X_1 and X_2 such that (1) X_1 joins g_1 and $(v)^{n-}$, (2) X_2 joins g_2 and $(s_1)^{n-}$, and (3) $X_1 \cup X_2$ spans $AQ_n^{0,1}$. We set

$$\begin{aligned} P_{2n-3} &= \langle u, g_1, X_1, (v)^{n-}, v \rangle, \text{ and} \\ P_{2n-2} &= \langle u, g_2, X_2, (s_1)^{n-}, s_1, \ldots, s_{j-1}, \\ &\quad (s_{j-1})^{n-1}, T, (v)^{n-1}, v \rangle. \end{aligned}$$

Subcase 2.2: $H = \langle (u)^n, S, (w_2)^n, v, (w_3)^n, \rangle$ where $S = \langle s_0 = (u)^n, s_1, \ldots, s_j = (w_2)^n \rangle$. The same idea considered in Subcase 2.1. We set

$$\begin{aligned} P_{2n-3} &= \langle u, g_1, X_1, (v)^{n-}, v \rangle, \text{ and} \\ P_{2n-2} &= \langle u, g_2, X_2, (s_1)^{n-}, s_1, \ldots, s_{j-1}, \\ &\quad (s_{j-1})^{n-1}, T, (v)^{n-1}, v \rangle. \end{aligned}$$

Then $\{P_1, P_2, \ldots, P_{2n-2}\}$ forms a $(2n-2)^*$-container of AQ_n between u and v.

Case III: $k = 2n-1$. By Lemma 2, let $\{(s_{j-1})^{n-1}, (v)^{n-1}\}$ and $\{(u)^{n-}, (v)^{(n-1)-}\}$ be two pairs of four distinct nodes of $AQ_n^{1,1}$ where s_{j-1} be the node considered in Case II. There exist two node-disjoint paths T_1 and T_2 such that (1) T_1 joins $(s_{j-1})^{n-1}$ and $(v)^{n-1}$, (2) T_2 joins $(u)^{n-}$ and $(v)^{(n-1)-}$, and (3) $T_1 \cup T_2$ spans $AQ_n^{1,1}$. Now we replace the path P_{2n-2} considered in Case 2 by P'_{2n-2} and P'_{2n-1}.

$$\begin{aligned} P'_{2n-2} &= \langle u, \ldots, (s_{j-1})^{n-1}, T_1, (v)^{n-1}, v \rangle, \text{ and} \\ P'_{2n-1} &= \langle u, (u)^{n-}, T_2, (v)^{(n-1)-}, v \rangle. \end{aligned}$$

Then $\{P_1, P_2, \ldots, P_{2n-3}, P'_{2n-2}, P'_{2n-1}\}$ forms a $(2n-1)^*$-container of AQ_n between u and v. $\square$

It is easy to check that AQ_1 and AQ_2 are super spanning connected. With Lemmas 6, 7, 8, and 9, we have the following result:

Theorem 4 *AQ_n is super spanning connected if and only if $n \neq 3$.*

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